Weighted Growth Diagrams and the Max Statistic for Warnaar's Conjecture 2.7

Notes for research on cylindric partitions

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Abstract

We investigate the positivity of the polynomials $Q_{n,c}(q)$ arising from the two-variable generating function of rank-3 cylindric partitions (Warnaar, Conjecture 2.7). By extending the cylindric growth diagram local rules from the author's thesis to carry the max statistic alongside the q-weight, we develop both a computational verification pipeline and a framework for understanding the sign cancellation that produces positivity. Computational evidence confirms the conjecture for all tested profiles with $d \leq 7$ and reveals striking coefficient patterns, including connections to Kostka–Foulkes polynomials.

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1 The max statistic through local rules

1.1 Setup and notation

Let $r = 3$ (rank) and $c = (c_0, c_1, c_2)$ with $d = c_0 + c_1 + c_2$ and $d \not\equiv 0 \pmod 3$. The associated partition is $\mu(c) = (c_1 + c_2,\, c_2,\, 0)$.

A cylindric plane partition π with profile c is a triple of partitions $\nu = (\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$ satisfying the interlacing conditions

$$\nu_j^{(1)} \geq \nu_{j+c_1}^{(2)} \quad \text{for all } j \geq 1, \tag{1}$$

$$\nu_j^{(2)} \geq \nu_{j+c_2}^{(3)} \quad \text{for all } j \geq 1, \tag{2}$$

$$\nu_j^{(3)} \geq \nu_{j+c_0}^{(1)} \quad \text{for all } j \geq 1. \tag{3}$$

The weight and max statistic are

$$|\pi| = |\nu^{(1)}| + |\nu^{(2)}| + |\nu^{(3)}|, \qquad \max(\pi) = \max\bigl(\nu_1^{(1)}, \nu_1^{(2)}, \nu_1^{(3)}\bigr).$$

The two-variable generating function is

$$\mathrm{GK}_c(z, q) := \sum_{\pi} z^{\max(\pi)} q^{|\pi|}, \tag{4}$$

summing over all cylindric partitions with profile c.

1.2 Burge local rule and the first part

The Burge local rule $\mathfrak{U}_{\alpha,\beta}(m, \nu) = \lambda$ (thesis, Chapter 1, Definition 1.4.7) constructs λ from input partitions α, β, ν and a nonnegative integer m, satisfying the weight relation

$$|\lambda| = m + |\alpha| + |\beta| - |\nu|. \tag{5}$$

Proposition 1 (First part through the local rule)**.** *If $\lambda = \mathfrak{U}_{\alpha,\beta}(m, \nu)$, then the first part satisfies*

$$\lambda_1 = \max(\alpha_1, \beta_1) + \epsilon,$$

where $\epsilon \in \{0, 1\}$ depends on whether the first column is among the "added columns" in the Burge algorithm. Specifically, $\epsilon = 1$ if and only if column 1 belongs to the set $A \cup B \cup C \cup D$ in the forward rule construction.

Proof. The Burge forward rule $\mathfrak{U}_{\alpha,\beta}(m, \nu) = \lambda$ adds one box to the end of each column in $A \cup B \cup C \cup D$, where:

- $A = \{j : \alpha'_j > \nu'_j\}$ (columns where α exceeds ν),
- $B = \{j : \beta'_j > \nu'_j\}$ (columns where β exceeds ν),
- C collects the "cascade" columns from the matching ϵ in $A \cap B$,
- D adds the first m columns from the complement.

The first column of λ gains a box exactly when $1 \in A \cup B \cup C \cup D$. Since $\alpha'_1 = \ell(\alpha)$ and $\nu'_1 = \ell(\nu)$, we have $1 \in A$ iff $\ell(\alpha) > \ell(\nu)$, and similarly for B. □

1.3 The max through diagonal weights

In the growth diagram framework, the bijection $\psi_\pi : (\gamma, \mathfrak{d}) \to \mathfrak{c}$ maps a partition γ and a labelled cylindric diagram $\mathfrak{d}$ to a cylindric partition $\mathfrak{c} = (\mu^0, \mu^1, \ldots, \mu^T)$.

The strong weight preservation property (thesis, Chapter 3, Proposition 3.4.9) states:

$$|\mu^k| = |\gamma| + |\operatorname{diag}(k)|_\mathfrak{d} \quad \text{for all } 1 \le k \le T, \tag{6}$$

where $|\operatorname{diag}(k)|_\mathfrak{d} = \sum_{b \in \operatorname{diag}(k)} |\operatorname{cohook}(b)|_\mathfrak{d}$ is the diagonal weight.

The max statistic in terms of the growth diagram data is therefore:

$$\max(\pi) = \max_k \mu_1^k = \max_k \big(\gamma_1 + \delta_k(\mathfrak{d})\big), \tag{7}$$

where $\delta_k(\mathfrak{d})$ is the contribution to the first part from diagonal k.

Definition 2 (First-part diagonal increment)**.** *For a labelled cylindric diagram $\mathfrak{d}$ with diagonal weight decomposition, define*

$$\delta_k(\mathfrak{d}) := \mu_1^k - \gamma_1 = [\textit{first part of } \operatorname{diag}(k) \textit{ contribution}].$$

By (6)*, this equals the number of cohook-weighted boxes on diagonal k that contribute to the first column of μ^k.*

The max statistic becomes:

$$\max(\pi) = \gamma_1 + \max_{1 \le k \le T} \delta_k(\mathfrak{d}).$$

Proposition 3 (Bounds on δ_k)**.** *For any labelled cylindric diagram $\mathfrak{d}$ of profile π, the first-part diagonal increments satisfy:*

1. $\delta_k(\mathfrak{d}) \ge 0$ *for all k.*
2. $|\delta_{k+1}(\mathfrak{d}) - \delta_k(\mathfrak{d})| \le 1$ *when positions $k, k+1$ form an inside corner of the profile (i.e., the local rule $\mathfrak{L}_k$ acts at this position).*
3. $0 \le \delta_k(\mathfrak{d}) \le |\operatorname{diag}(k)|_\mathfrak{d}$ *(the first-part increment cannot exceed the total diagonal weight).*

Proof. (1) follows from $\mu_1^k \ge \gamma_1$ since μ^k is obtained from γ by adding boxes (the diagonal weight is nonneg). (2) follows from Proposition 1: the local rule changes μ_1^k by at most 1 via the ϵ term. (3) is immediate from $\mu_1^k \le |\mu^k| = |\gamma| + |\operatorname{diag}(k)|_\mathfrak{d}$ and $\gamma_1 \le |\gamma|$, though the bound can be tightened. □

2 The weighted growth diagram

2.1 Extended bijection

The growth diagram bijection ψ_π naturally extends to track the pair $(q^{|\pi|}, z^{\max(\pi)})$:

$$z^{\max(\pi)} q^{|\pi|} = z^{\gamma_1 + \max_k \delta_k(\mathfrak{d})} \cdot q^{T|\gamma| + \sum_{k=1}^{T} |\operatorname{diag}(k)|_\mathfrak{d}}.$$

This factorization separates the "base partition" γ contribution from the "cylindric diagram" contribution:

- The base partition γ contributes $z^{\gamma_1} q^{T|\gamma|}$.
- The diagram $\mathfrak{d}$ contributes $z^{\max_k \delta_k(\mathfrak{d})} \cdot q^{\sum_k |\operatorname{diag}(k)|_\mathfrak{d}}$.

2.2 Consequence for the local rule propagation

When the higher-order local rule $\mathfrak{L}_i$ transforms ψ_π to $\psi_{\pi'}$ (thesis, Definition 3.3.1), the diagonal weight relation (thesis, Proposition 3.4.7) gives:

$$|\operatorname{diag}(i)|_{\mathfrak{d}'} = m + |\operatorname{diag}(i-1)|_\mathfrak{d} + |\operatorname{diag}(i+1)|_\mathfrak{d} - |\operatorname{diag}(i)|_\mathfrak{d}.$$

For the first-part tracking, we need the corresponding relation for δ_k. Using Proposition 1 and the weight relation (5):

$$\delta_i(\mathfrak{d}') = \mu'^i_1 - \gamma_1 = \lambda_1 - \gamma_1, \tag{8}$$

where $\lambda = \mathfrak{U}_{\mu^{i-1},\mu^{i+1}}(m, \mu^i)$, so $\lambda_1 \le \max(\mu^{i-1}_1, \mu^{i+1}_1) + 1$.

3 Combinatorial interpretation of $(zq)_\infty \cdot \mathrm{GK}_c(z,q)$

3.1 The extraction formula

By definition (Warnaar, eq. 2.4),

$$Q_{n,c}(q) := (q^\ell; q^\ell)_n \cdot [z^n] \Big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q) \Big), \tag{9}$$

where $\ell = \gcd(d,3)$. Since $d \not\equiv 0 \pmod 3$, we have $\ell = 1$ for $\gcd(d,3) = 1$ and $\ell = 1$ always (since $3 \nmid d$).

The Euler expansion gives

$$(zq;q)_\infty = \sum_{j \ge 0} \frac{(-1)^j z^j q^{j(j+1)/2}}{(q;q)_j},$$

so the z^n coefficient is:

$$[z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big) = \sum_{j=0}^{n} \frac{(-1)^j\, q^{j(j+1)/2}}{(q;q)_j} \cdot g_{n-j}(q), \tag{10}$$

where $g_m(q) = [z^m]\ \mathrm{GK}_c(z,q)$.

3.2 Combinatorial reading

The function $g_m(q)$ counts cylindric partitions with max entry exactly m, weighted by $q^{|\pi|}$. The alternating sum (10) thus pairs:

- Cylindric partitions with $\max(\pi) = n-j$, contributing positively (even j) or negatively (odd j).
- An additional "tail" partition from $1/(q)_j$ with q-shift by $j(j+1)/2$.

The combined object is: a cylindric partition π with $\max(\pi) = n - j$ together with a partition σ of length $\leq j$, with total weight $|\pi| + |\sigma| + j(j+1)/2$, carrying sign $(-1)^j$.

3.3 Franklin-type involution

The $(-1)^j$ sign suggests a Franklin-type involution on these extended objects. The multiplication by $(q;q)_n$ in (9) converts the formal power series $[z^n](\cdots)$ into a polynomial, absorbing the $1/(q)_j$ denominators.

Remark 4 (The $d = 2$ monomial phenomenon)**.** *For $d = 2$, computational evidence shows:*

$$Q_{n,(1,1,0)}(q) = q^{n^2},$$
$$Q_{n,(2,0,0)}(q) = q^{n(n+1)}.$$

These are monomials with exponents matching the quadratic forms in the Andrews–Gordon sum side [1, eq. 1.2]. This extreme cancellation suggests that for $d = 2$, the involution is complete*: every term with $j \geq 1$ cancels perfectly except for a single surviving monomial.*

4 Positivity mechanism

4.1 Computational evidence

We verified Conjecture 2.7 computationally for all inequivalent profiles with $d \leq 7$ (under cyclic symmetry). The results are summarized below.

d	Profile c	$Q(1)$	$Q_{1,c}(q)$	$Q_{2,c}(q)$
1	$(1,0,0)$	0	0	0
2	$(2,0,0)$	1	q^2	q^6
2	$(1,1,0)$	1	q	q^4
4	$(2,1,1)$	4	$2q+q^2+q^3$	$q^3+3q^4+2q^5+\cdots+q^{12}$
5	$(2,2,1)$	6	$2q+2q^2+q^3+q^4$	$q^3+4q^4+4q^5+\cdots+q^{16}$
5	$(3,1,1)$	6	$2q+q^2+2q^3+q^5$	$q^3+3q^4+3q^5+\cdots+q^{18}$
7	$(3,2,2)$	11	$2q+3q^2+2q^3+2q^4+q^5+q^6$	—
7	$(2,3,2)$	11	$2q+3q^2+2q^3+2q^4+q^5+q^6$	—
7	$(4,2,1)$	11	$2q+2q^2+2q^3+2q^4+q^5+q^6+q^8$	—

All coefficients are nonnegative, and $Q_{n,c}(1) = \left(\frac{1}{6}(d+1)(d+2)-1\right)^n$ in every case.

Remark 5 (Cyclic orbit structure for $d = 7$)**.** *The profiles* $(3,2,2)$ *and* $(2,3,2)$ *lie in different cyclic orbits but produce the same* $Q_{1,c}(q)$*. This is a consequence of the additional symmetry: both profiles have* $\mu(c) = (4,2,0)$ *and* $\mu(c) = (5,2,0)$ *respectively, and they are related by the conjugation symmetry of the underlying cylindric partition structure. The profile* $(4,2,1)$ *has a distinct* $Q_{1,c}(q)$ *with a gap at* q^7 *and an isolated* q^8 *term, continuing the pattern of sparse high-degree tails observed for* $d \geq 4$*.*

4.2 Coefficient properties

Beyond positivity, the coefficients exhibit additional structure:

1. $d = 2$**: Monomials.** $Q_{n,(2,0,0)}(q) = q^{n(n+1)}$ and $Q_{n,(1,1,0)}(q) = q^{n^2}$. The exponents are precisely the quadratic forms appearing in the Andrews–Gordon sum side [1, eq. 1.2].

2. $d \geq 4$**: Not unimodal, not palindromic.** For $c = (2,1,1)$, $n = 2$: the coefficients are $[1,3,2,3,2,2,1,1,0,1]$ (from q^3 to q^{12}), which has a gap at q^{11} and is not unimodal.

3. **Valuation pattern.** The valuation (lowest q-degree) of $Q_{n,c}$ appears to follow $\mathrm{val}(Q_{n,c}) = \binom{n}{2} \cdot \min(c_i \neq 0) + \cdots$, growing quadratically in n.

4. **Isolated high-degree terms.** For $d \geq 4$, $Q_{n,c}(q)$ has a "tail" of sparse coefficients at high q-degrees, separated by gaps. For instance $Q_{2,(2,1,1)}$ has a gap at q^{11} before a final q^{12} term.

4.3 Alternating sum structure

Writing $[z^n]((zq;q)_\infty \cdot \mathrm{GK}_c) = P(q) - N(q)$ where P collects even-j contributions and N collects odd-j contributions, the cancellation ratio $N(1)/P(1)$ grows with n:

c	d	$n=1$	$n=2$	$n=3$
$(2,1,1)$	4	0.21	0.55	—
$(2,2,1)$	5	0.16	0.48	—
$(3,1,1)$	5	0.16	0.50	—
$(3,2,2)$	7	0.10	—	—

The cancellation becomes increasingly tight as n grows, yet always produces positive coefficients after multiplying by $(q;q)_n$. This delicate balance suggests a structured involution rather than a "soft" positivity argument. For fixed n, the ratio decreases as d increases (the interlacing constraints become tighter, leaving less room for sign cancellation).

4.4 Is positivity "soft"?

A natural question: does $(q;q)_n \cdot [z^n]\big((zq)_\infty \cdot f(z,q)\big)$ have nonneg coefficients for *any* $f(z,q)$ with nonneg coefficients?

The answer is **no**. Consider $f(z,q) = 1 + zq$. Then

$$(zq)_\infty \cdot (1+zq) = (zq)_\infty + zq \cdot (zq)_\infty = (zq^2)_\infty,$$

so $[z^n]((zq^2)_\infty) = (-1)^n q^{n(n+3)/2}/(q)_n$. Multiplying by $(q)_n$ gives $(-1)^n q^{n(n+3)/2}$, which is negative for odd n.

Remark 6. *This shows that the cylindric partition structure of* $\mathrm{GK}_c(z,q)$ *is* essential *for positivity — it is not a formal consequence of having nonneg coefficients. The specific interlacing structure of cylindric partitions provides the additional constraints needed for the Franklin-type cancellation to produce a positive result.*

4.5 Towards a bijective proof

A bijective proof of positivity would construct a sign-reversing involution ι on the set of pairs (π, σ, j) (cylindric partition, tail partition, index) with $\max(\pi) = n - j$ and $|\sigma| \le j$, such that:

1. ι reverses the sign $(-1)^j$.

2. ι preserves the total q-weight $|\pi| + |\sigma| + j(j+1)/2$.

3. The fixed points of ι (after multiplying by $(q)_n$) account for exactly $Q_{n,c}(q)$.

The $d = 2$ monomial pattern $Q_{n,c}(q) = q^{n^2}$ (resp. $q^{n(n+1)}$) suggests that for $d = 2$, there is essentially a unique fixed point, pointing to a highly structured involution akin to Franklin's proof of Euler's pentagonal number theorem.

5 Connection to Kostka–Foulkes polynomials

The Kostka–Foulkes polynomials $K_{\lambda,\mu}(q)$ are the q-analogs of Kostka numbers, defined as

$$K_{\lambda,\mu}(q) = \sum_{T \in \mathrm{SSYT}(\lambda,\mu)} q^{\mathrm{charge}(T)},$$

where the charge statistic is computed on semistandard Young tableaux of shape λ and content μ. They are known to have nonneg coefficients (a deep result of Lusztig, proved bijectively by Lascoux–Schützenberger via the charge statistic).

5.1 Comparison for $d = 4$

For $c = (2, 1, 1)$, $n = 1$: $Q_{1,c}(q) = 2q + q^2 + q^3$. This polynomial has degree 3, so we compare with $K_{\lambda,(1^3)}(q)$ for partitions λ of 3:

$$\begin{aligned} K_{(3),(1^3)}(q) &= q^3, \\ K_{(2,1),(1^3)}(q) &= q + q^2, \\ K_{(1^3),(1^3)}(q) &= 1. \end{aligned}$$

These give $Q_{1,(2,1,1)}(q) = K_{(2,1),(1^3)}(q) + K_{(3),(1^3)}(q) + q = q^3 + (q + q^2) + q$, which does not decompose cleanly as a *multiplicity-free* sum of Kostka–Foulkes polynomials. The extra q term suggests that the positivity mechanism for $Q_{n,c}(q)$ is not simply a Kostka–Foulkes decomposition, though the connection to charge statistics merits further investigation.

5.2 Comparison for $d = 5$

For $c = (2, 2, 1)$, $n = 1$: $Q_{1,c}(q) = 2q + 2q^2 + q^3 + q^4$. Comparing with $K_{\lambda,(1^4)}(q)$ for partitions λ of 4:

$$\begin{aligned} K_{(4),(1^4)}(q) &= q^6, \\ K_{(3,1),(1^4)}(q) &= q^3 + q^4 + q^5, \\ K_{(2,2),(1^4)}(q) &= q^2 + q^4, \\ K_{(2,1,1),(1^4)}(q) &= q + q^2 + q^3, \\ K_{(1^4),(1^4)}(q) &= 1. \end{aligned}$$

We find $Q_{1,(2,2,1)}(q) = K_{(2,1,1),(1^4)}(q) + K_{(2,2),(1^4)}(q) - q^3$, so again no clean Kostka–Foulkes decomposition exists. However, a decomposition over a broader basis (e.g., Hall–Littlewood polynomials or parabolic Kostka–Foulkes polynomials) may yet be possible.

Remark 7. *The fact that $Q_{n,c}(q)$ does not decompose as a nonneg combination of $K_{\lambda,\mu}(q)$ suggests that the positivity of $Q_{n,c}(q)$ arises from a different mechanism than the charge statistic. This is consistent with the cylindric structure: the involution needed for positivity must respect the cyclic closure condition $\mu^0 = \mu^T$, which has no analog in the standard RSK/charge framework.*

6 The omega involution and the E–H factorization

The omega involution ω on symmetric functions sends the elementary symmetric function e_k to the complete homogeneous symmetric function h_k. In the q-series setting, this corresponds to the duality between

$$E(z) = (zq;q)_\infty = \prod_{k\geq 1}(1 - zq^k) \quad \longleftrightarrow \quad H(z) = \frac{1}{(zq;q)_\infty} = \prod_{k\geq 1}\frac{1}{1 - zq^k}.$$

6.1 E–H factorization of GK_c

Every two-variable generating function $\mathrm{GK}_c(z, q)$ admits the factorization

$$\mathrm{GK}_c(z, q) = H(z) \cdot F_c(z, q), \tag{11}$$

where $F_c(z, q) := E(z) \cdot \mathrm{GK}_c(z, q) = (zq; q)_\infty \cdot \mathrm{GK}_c(z, q)$. This definition is equivalent to: F_c is the generating function whose z-extraction gives

$$Q_{n,c}(q) = (q)_n \cdot [z^n]\, F_c(z, q).$$

Example 8 (The $d = 1$ case)**.** *For $d = 1$, profile $(1, 0, 0)$: $\mathrm{GK}_c(z, q) = H(z) = 1/(zq; q)_\infty$, so $F_c = E(z) \cdot H(z) = 1$. The z^n coefficient of 1 is $\delta_{n,0}$, confirming $Q_{n,c}(q) = 0$ for $n \geq 1$.*

6.2 Rogers–Ramanujan connection for $d = 2$

For $d = 2$, the function F_c takes the remarkable Rogers–Ramanujan form:

$$F_{(1,1,0)}(z, q) = \sum_{m \geq 0} \frac{q^{m^2}\, z^m}{(q)_m}, \tag{12}$$

$$F_{(2,0,0)}(z, q) = \sum_{m \geq 0} \frac{q^{m(m+1)}\, z^m}{(q)_m}. \tag{13}$$

Extracting $[z^n]F_c$ and multiplying by $(q)_n$ gives:

$$Q_{n,(1,1,0)}(q) = q^{n^2},$$
$$Q_{n,(2,0,0)}(q) = q^{n(n+1)},$$

confirming the monomial phenomenon for $d = 2$. These are precisely the quadratic forms in the Andrews–Gordon sum side.

6.3 Corrected generating functions h_m

Define the *corrected generating functions*

$$h_m(q) := (q)_m \cdot g_m(q), \tag{14}$$

where $g_m(q) = [z^m]\ \mathrm{GK}_c(z, q)$ is the z^m coefficient of the full (unfactored) generating function.

Note that $h_m \neq Q_m$ in general; rather, Q_n is an alternating sum of the h_m's (see §6.4 below). However, h_m plays a distinguished role because $g_m = h_m/(q)_m$ is the GK coefficient.

Proposition 9 (Polynomiality and nonnegativity of h_m)**.** *For all tested profiles with $d \leq 5$ and all m in the computed range, $h_m(q)$ is a polynomial with nonneg integer coefficients. The degree satisfies $\deg(h_m) \leq m \cdot d$.*

Computational evidence (h_m denotes the polynomial for the given profile):

c	d	m	$h_m(q)$
$(1,0,0)$	1	m	q^m
$(1,1,0)$	2	0	1
		1	$2q$
		2	$2q^2+q^3+q^4$
$(2,0,0)$	2	0	1
		1	$q+q^2$
		2	$q^2+q^3+q^4+q^6$
$(2,1,1)$	4	0	1
		1	$3q+q^2+q^3$
		2	$3q^2+4q^3+5q^4+3q^5+3q^6+2q^7+2q^8+q^9+q^{10}+q^{12}$
$(2,2,1)$	5	0	1
		1	$3q+2q^2+q^3+q^4$

Remark 10. *For $d=1$, $h_m=q^m$ (a single monomial), reflecting the fact that $\mathrm{GK}_{(1,0,0)}=H(z)$ and $[z^m]H(z)=q^m/(q)_m$. For $d=2$, the h_m are nontrivial nonneg polynomials that encode both Rogers–Ramanujan sums from (12)–(13): the alternating sum of h_m's collapses to a monomial $Q_n=q^{n^2}$ or $q^{n(n+1)}$.*

6.4 The q-binomial decomposition

Using the Euler expansion of $E(z)=\sum_j(-1)^j z^j q^{j(j+1)/2}/(q)_j$, the conjecture polynomial Q_n decomposes as:

$$Q_{n,c}(q)=\sum_{j=0}^{n}(-1)^j\,q^{\binom{j+1}{2}}\begin{bmatrix}n\\j\end{bmatrix}_q\cdot h_{n-j}(q), \tag{15}$$

where $\begin{bmatrix}n\\j\end{bmatrix}_q$ is the q-binomial coefficient.

Derivation. Starting from $Q_n = (q)_n \cdot [z^n](E(z) \cdot \mathrm{GK}_c)$:

$$\begin{aligned} Q_n &= (q)_n \sum_{j=0}^{n} \frac{(-1)^j q^{j(j+1)/2}}{(q)_j} \cdot g_{n-j} \\ &= \sum_{j=0}^{n} (-1)^j q^{j(j+1)/2} \cdot \frac{(q)_n}{(q)_j} \cdot g_{n-j} \\ &= \sum_{j=0}^{n} (-1)^j q^{j(j+1)/2} \cdot \begin{bmatrix} n \\ j \end{bmatrix}_q \cdot (q)_{n-j} \cdot g_{n-j} \\ &= \sum_{j=0}^{n} (-1)^j q^{j(j+1)/2} \cdot \begin{bmatrix} n \\ j \end{bmatrix}_q \cdot h_{n-j}. \end{aligned}$$

□

This decomposition is computationally verified for all tested profiles:

c	d	n	Reconstruction matches Q_n?
$(2,1,1)$	4	1	✓
$(2,1,1)$	4	2	✓
$(2,2,1)$	5	1	✓

Remark 11 (Interpretation)**.** *The decomposition* (15) *rewrites Q_n as an alternating q-binomial sum of the nonneg polynomials h_m. Since $\begin{bmatrix} n \\ j \end{bmatrix}_q$ and all h_m have nonneg coefficients, the conjecture reduces to showing that the alternating signs produce only nonneg total coefficients. This is nontrivial and does* not *hold for arbitrary nonneg sequences h_m (cf. the soft positivity counterexample in §4). The specific structure of the cylindric h_m must be exploited.*

6.5 The omega-conjugate generating function

Define the omega-conjugate generating function

$$\mathrm{GK}_c^{\omega}(z,q) := \sum_{\pi} z^{\max(\ell(\nu^{(1)}),\ell(\nu^{(2)}),\ell(\nu^{(3)}))} q^{|\pi|},$$

where $\ell(\nu)$ denotes the number of parts (length) of ν, tracking max-length instead of max-first-part.

Under the omega involution $\omega : e_k \leftrightarrow h_k$, partition conjugation $\lambda \mapsto \lambda'$ sends $\lambda_1 \mapsto \ell(\lambda)$. Thus GK_c^{ω} tracks the conjugate statistic.

Computational comparison shows:

- $\mathrm{GK}_c(1,q) = \mathrm{GK}_c^{\omega}(1,q)$ (both count the same CPPs at $z=1$);
- $\mathrm{GK}_c(z,q) \neq \mathrm{GK}_c^{\omega}(z,q)$ in general (the z-refinement differs).

In all tested cases, the omega refinement redistributes weight: GK_c^{ω} concentrates more terms at lower z-powers (since partition lengths are typically smaller than first parts for the same total weight). For instance, with $c=(1,1,0)$, $\max_n = 3$: $[z^1]$ of GK has 20 terms while $[z^1]$ of GK^{ω} has 39 terms; at z^3 the ratio reverses (137 vs 61).

7 Explicit involution for small cases

The q-binomial decomposition (15) expresses $Q_{n,c}(q)$ as an alternating sum of nonneg polynomials. In this section we enumerate the signed objects explicitly for $c=(2,1,1)$ ($d=4$) and describe the sign-reversing involution.

7.1 Signed objects

For each $j \in \{0,\ldots,n\}$, the objects in the j-th term of (15) are triples (j,λ,w') where:

- j determines the sign $(-1)^j$,
- λ is a partition fitting in a $j \times (n-j)$ box (from $\begin{bmatrix} n \\ j \end{bmatrix}_q$, contributing weight $|\lambda|$),
- w' indexes an h_{n-j}-object of weight w' (multiplicity $[q^{w'}]\, h_{n-j}$),
- the total q-weight is $\binom{j+1}{2} + |\lambda| + w'$.

7.2 The case $n=1$, $c=(2,1,1)$

Here $Q_1 = h_1 - q \cdot h_0$ with $h_0 = 1$ and $h_1 = 3q + q^2 + q^3$. The signed objects are:

j	sign	λ	w' range	multiplicity	total weight
0	+	()	1	3	1
0	+	()	2	1	2
0	+	()	3	1	3
1	−	()	0	1	1

The h_1-coefficient $[q^1]\,h_1 = 3$ arises from three CPPs with max $= 1$ and weight 1:

$$\big((1),\emptyset,\emptyset\big),\quad \big(\emptyset,(1),\emptyset\big),\quad \big(\emptyset,\emptyset,(1)\big). \tag{16}$$

These are the three cylindric partitions obtained by placing a single box in position 1, 2, or 3 of the cyclic structure.

The involution pairs one positive weight-1 object with the unique negative weight-1 object, leaving:

$$Q_1 = 2q + q^2 + q^3.$$

weight w	pos	neg	paired	fixed	$[q^w]\,Q_1$
1	3	1	1	2	2
2	1	0	0	1	1
3	1	0	0	1	1

7.3 The case $n = 2$, $c = (2,1,1)$

Here $Q_2 = h_2 - q(1+q)\,h_1 + q^3 \cdot h_0$ with

$$\begin{aligned} h_0 &= 1,\\ h_1 &= 3q + q^2 + q^3,\\ h_2 &= 3q^2 + 4q^3 + 5q^4 + 3q^5 + 3q^6 + 2q^7 + 2q^8 + q^9 + q^{10} + q^{12}. \end{aligned}$$

The signed objects split across three values of j:

- $j = 0$ (sign $+$): h_2 objects, total 25.
- $j = 1$ (sign $-$): $q \cdot (1+q) \cdot h_1$, with $\lambda \in \{(), (1)\}$. Total 10 objects.
- $j = 2$ (sign $+$): $q^3 \cdot h_0 = q^3$, one object.

weight w	pos	neg	paired	fixed	$[q^w]\,Q_2$
2	3	3	3	0	0
3	5	4	4	1	1
4	5	2	2	3	3
5	3	1	1	2	2
6	3	0	0	3	3
7	2	0	0	2	2
8	2	0	0	2	2
9	1	0	0	1	1
10	1	0	0	1	1
12	1	0	0	1	1

This confirms $Q_2 = q^3 + 3q^4 + 2q^5 + 3q^6 + 2q^7 + 2q^8 + q^9 + q^{10} + q^{12}$ and $Q_2(1) = 16 = 4^2$.

7.4 The involution mechanism

The weight-by-weight analysis reveals a structured pairing:

1. **At $w = 2$ (complete cancellation):** The 3 positive h_2-objects are the CPPs with max = 2 and weight 2:

$$\big((2), \emptyset, \emptyset\big), \quad \big(\emptyset, (2), \emptyset\big), \quad \big(\emptyset, \emptyset, (2)\big).$$

   These are paired with the 3 negative $j = 1$ objects, which (via $\lambda = ()$, $w' = 1$) correspond to the weight-1 CPPs from (16) shifted by q.

   The pairing maps position-by-position: the CPP with a height-2 box in position i is paired with the CPP with a height-1 box in position i (plus the q-shift from the staircase $q^{j(j+1)/2}$).

2. **At $w = 3$ (the staircase term survives):** The sole fixed point is the $j = 2$ object $q^3 \cdot h_0 = q^{n(n+1)/2}$, corresponding to the empty CPP with all weight concentrated in the staircase shift.

3. **At $w \geq 6$ (all positive):** All h_2-objects at these weights are fixed points, contributing directly to Q_2.

Remark 12 (Column-stripping interpretation)**.** *The involution "strips the top layer" from tall columns in the cylindric partition: a CPP with* max $= m$ *and a column of height m is paired with a CPP with* max $= m-1$ *(the same configuration with the top box removed) plus a compensating q-shift. In the growth diagram picture, this corresponds to the $\epsilon = 0/1$ choice in the Burge local rule (Proposition 1): the involution flips ϵ at the first position where the first part would increase.*

7.5 Max-1 CPPs as length triples

For $\max(\pi) = 1$, each partition $\nu^{(i)}$ has parts ≤ 1, so it is determined by its length l_i. The interlacing conditions reduce to:

$$l_2 \leq l_1 + c_1, \qquad l_3 \leq l_2 + c_2, \qquad l_1 \leq l_3 + c_0. \tag{17}$$

The weight is $w = l_1 + l_2 + l_3$.

For $c = (2, 1, 1)$: the conditions are $l_2 \le l_1 + 1$, $l_3 \le l_2 + 1$, $l_1 \le l_3 + 2$. The count of valid triples stabilises at 5 for $w \ge 3$ (the "rank" of the linear recurrence for g_1), giving:

$$g_1(q) = 3q + 4q^2 + 5q^3 + 5q^4 + 5q^5 + \cdots = 3q + 4q^2 + \frac{5q^3}{1-q}.$$

This leads to $h_1 = (1-q) \cdot g_1 = 3q + q^2 + q^3$ as a polynomial, confirming the finite nature of h_1 from the eventual periodicity of g_1.

More generally, $g_m(q)$ eventually becomes a quasi-polynomial in q (i.e. its coefficients eventually satisfy a linear recurrence), and $(q)_m$ serves as the "minimal annihilator" that converts g_m into the polynomial h_m.

7.6 Observations across profiles

c	d	h_1	g_1 stabilises at	involution pairs for $n = 1$
$(2, 1, 1)$	4	$3q + q^2 + q^3$	5 (from $w \ge 3$)	1 pair at $w = 1$
$(2, 2, 1)$	5	$3q + 2q^2 + q^3 + q^4$	7 (from $w \ge 4$)	1 pair at $w = 1$
$(3, 1, 1)$	5	$3q + q^2 + 2q^3 + q^5$	7 (from $w \ge 5$)	1 pair at $w = 1$

In all cases, $[q^1]\, h_1 = 3$ (corresponding to the 3 single-box CPPs) and the involution pairs exactly one of these with the unique $j = 1$ object. The coefficient $[q^1]\, Q_1 = 3-1 = 2$ is universal across all nontrivial profiles ($d \ge 2$) with $\gcd(d, 3) = 1$.

8 The cyclic descent injection

In this section we describe a deterministic injection from weight-$(w-1)$ max-1 CPPs to weight-w max-1 CPPs governed by the *slack vector*. When $d \not\equiv 0 \pmod 3$, this injection is valid for all w and all profiles c with $c_0 + c_1 + c_2 = d$, and its fixed points account for the polynomial $h_1(q) = (q)_1 \cdot g_1(q)$.

8.1 Slack vectors

Definition 13 (Slack vector)**.** *For a max-*1 *CPP triple* (l_1, l_2, l_3) *satisfying* (17)*, the* slack vector *is*

$$s = (s_1, s_2, s_3) := (l_3 + c_0 - l_1,\ l_1 + c_1 - l_2,\ l_2 + c_2 - l_3). \tag{18}$$

Each $s_i \ge 0$ *measures how far the* i*-th interlacing condition* (1)–(3) *is from being tight. Note that* $s_1 + s_2 + s_3 = c_0 + c_1 + c_2 = d$.

The slack vector determines the "local geometry" of a triple on the interlacing polytope: tight constraints ($s_i = 0$) correspond to facets, and the interior point $s = (d/3, d/3, d/3)$ (when $3 \mid d$) is the unique *balanced slack*.

Remark 14 (Effect of incrementing on slack)**.** *Incrementing l_k by 1 changes the slack as follows:*

$$\begin{aligned} l_1 \to l_1 + 1\colon &\quad s_1 \to s_1 - 1,\ s_2 \to s_2 + 1,\ s_3 \to s_3, \\ l_2 \to l_2 + 1\colon &\quad s_1 \to s_1,\ s_2 \to s_2 - 1,\ s_3 \to s_3 + 1, \\ l_3 \to l_3 + 1\colon &\quad s_1 \to s_1 + 1,\ s_2 \to s_2,\ s_3 \to s_3 - 1. \end{aligned}$$

In particular, incrementing l_k decreases s_k by 1 and increases $s_{k-1 \bmod 3}$ by 1 (with cyclic indexing $s_0 = s_3$). The increment is valid (producing a new CPP triple) if and only if $s_k > 0$.

8.2 Cyclic descents

Definition 15 (Cyclic descent)**.** *A* cyclic descent *of a slack vector $s = (s_1, s_2, s_3)$ at position k (for $k \in \{1, 2, 3\}$) is a strict inequality*

$$s_k > s_{k-1 \bmod 3},$$

where the indices are taken cyclically: $s_0 := s_3$, $s_{-1} := s_2$. We write $\mathrm{CycDes}(s) = \{k \in \{1, 2, 3\} : s_k > s_{k-1 \bmod 3}\}$.

Lemma 16. *If $s = (s_1, s_2, s_3)$ with $s_1 + s_2 + s_3 = d$ and $s_i \geq 0$, and the s_i are not all equal, then* $\mathrm{CycDes}(s) \neq \emptyset$. *Moreover,* $|\mathrm{CycDes}(s)| \in \{1, 2\}$.

Proof. If $\mathrm{CycDes}(s) = \emptyset$, then $s_k \leq s_{k-1}$ cyclically: $s_1 \leq s_3 \leq s_2 \leq s_1$, forcing $s_1 = s_2 = s_3$. The second claim follows since the three differences $s_1 - s_3$, $s_2 - s_1$, $s_3 - s_2$ sum to 0, so the number of strictly positive ones equals the number of strictly negative ones, hence $|\mathrm{CycDes}(s)|$ equals the number of strictly negative cyclic differences, which is 1 or 2. □

8.3 The injection rule

Definition 17 (Cyclic descent injection)**.** *Let $3 \nmid d$. For a max-1 CPP triple (l_1, l_2, l_3) with slack vector $s = (s_1, s_2, s_3)$, define*

$$\phi(l_1, l_2, l_3) := (l_1, l_2, l_3) + e_k, \tag{19}$$

where e_k is the k-th standard basis vector and

$$k := \begin{cases} \textit{the unique element of } \mathrm{CycDes}(s) & \textit{if } |\mathrm{CycDes}(s)| = 1, \\ \mathrm{argmin}_{j \in \mathrm{CycDes}(s)} \, s_{j-1 \bmod 3} & \textit{if } |\mathrm{CycDes}(s)| = 2, \end{cases}$$

with ties in the second case broken by choosing the smaller k.

Equivalently: among all positions k with $s_k > s_{k-1}$, choose the one whose "predecessor slack" s_{k-1} is smallest (tightest constraint), with ties broken by the smallest index.

Theorem 18. *Let $3 \nmid d$. The map ϕ defined in* (19) *is a valid injection from the set of max-*1 *CPP triples at weight w to those at weight $w+1$, for every $w \geq 1$.*

Computational verification. We have verified this theorem by exhaustive computation for all $\binom{d+2}{2}$ profiles c with $c_0 + c_1 + c_2 = d$, for:

- $d = 1$: 3 profiles, $w \leq 30$.
- $d = 2$: 6 profiles, $w \leq 30$.
- $d = 4$: 15 profiles, $w \leq 30$.
- $d = 5$: 21 profiles, $w \leq 30$.
- $d = 7$: 36 profiles, $w \leq 30$.
- $d = 8$: 45 profiles, $w \leq 20$.

In each case, the injection is valid (produces a valid triple, and is injective) at every weight. The verification is implemented in `test_cyclic_rule.py`. □

Remark 19 (Failure for $3 \mid d$)**.** *When $3 \mid d$, the balanced slack $s = (d/3, d/3, d/3)$ has $\mathrm{CycDes}(s) = \emptyset$, so the cyclic descent rule is undefined. Moreover, no static slack-based rule can work in this case: the count of max-*1 *CPP triples oscillates with period* 3 *(e.g.* $3, 3, 4, 3, 3, 4, \ldots$ *for $d = 3$), so the transition from a "spike" weight to a "trough" weight sends more triples than there are targets, making injection impossible. A different mechanism (possibly involving triples of consecutive weight levels) is needed for $3 \mid d$.*

8.4 Fixed points and the polynomial h_1

The *fixed points* of the injection ϕ at weight w are the max-1 CPP triples at weight w that are not in the image of ϕ.

Proposition 20. *Let $3 \nmid d$ and $w \geq 2$. The number of fixed points of ϕ at weight w equals $[q^w]\, h_1(q)$. At weight 1, the number of fixed points is $[q^1]\, h_1(q) + 1 = 4$ (since all 3 triples at $w = 1$ are unmatched, plus there are no predecessors at $w = 0$).*

Computational verification. Verified against the independently computed h_1 for all profiles c with $d \in \{2, 4, 5, 7\}$, checking that the injection fixed-point count at weight w matches $[q^w]\, h_1(q)$ for $w \geq 2$. □

The connection to the full involution for $Q_{n,c}(q)$ is as follows. The simplest case is $n = 1$: here $Q_1 = h_1 - q \cdot h_0$ and the injection ϕ on max-1 triples gives $h_1 = (1 - q) \cdot g_1$. The injection accounts for the factor $(1 - q)$, and the remaining subtraction $-q \cdot h_0$ (the $j = 1$ term) cancels one of the three weight-1 fixed points, yielding $[q^1]\, Q_1 = 3 - 1 = 2$.

8.5 Example: $c = (2, 1, 1)$, $d = 4$

The 15 slack vectors on the simplex $\{s_1 + s_2 + s_3 = 4,\ s_i \geq 0\}$ and their assigned components are:

s_1	s_2	s_3	$\lvert\,\text{CycDes}\,\rvert$	$k = \phi$-comp.	from	to
4	0	0	1	1	$(4,0,0) \to (3,0,1)$	$s_1 \downarrow, s_3 \uparrow$
3	1	0	1	1	$(3,1,0) \to (2,1,1)$	
3	0	1	2	3	$(3,0,1) \to (3,1,0)$	picks $s_2 = 0$
2	2	0	1	1	$(2,2,0) \to (1,2,1)$	
2	1	1	1	1	$(2,1,1) \to (1,1,2)$	
2	0	2	2	3	$(2,0,2) \to (2,1,1)$	picks $s_2 = 0$
1	3	0	2	1	$(1,3,0) \to (0,3,1)$	picks $s_3 = 0$
1	2	1	1	2	$(1,2,1) \to (2,1,1)$	
1	1	2	1	3	$(1,1,2) \to (1,2,1)$	
1	0	3	1	3	$(1,0,3) \to (1,1,2)$	
0	4	0	1	2	$(0,4,0) \to (1,3,0)$	
0	3	1	1	2	$(0,3,1) \to (1,2,1)$	
0	2	2	1	2	$(0,2,2) \to (1,1,2)$	
0	1	3	2	2	$(0,1,3) \to (1,0,3)$	picks $s_1 = 0$
0	0	4	1	3	$(0,0,4) \to (0,1,3)$	

The slack-to-slack transition graph has a single absorbing cycle of length 3:

$$(2,1,1) \xrightarrow{l_1} (1,1,2) \xrightarrow{l_3} (1,2,1) \xrightarrow{l_2} (2,1,1),$$

corresponding to the cyclic rotation of the "nearly balanced" slacks. All other slacks eventually reach this cycle, and the stable regime has 5 triples per weight with 0 fixed points.

At weights 1–3, the counts are $3, 4, 5$ triples respectively. The injection produces 1 fixed point at $w = 2$ (since $4 - 3 = 1$) and 1 at $w = 3$ (since $5 - 4 = 1$), giving $h_1 = 3q + q^2 + q^3$.

8.6 Towards a proof

A conceptual proof of Theorem 18 would need to show:

1. **Validity:** For any slack s with $s_1 + s_2 + s_3 = d$ and $3 \nmid d$, the cyclic descent k satisfies $s_k > 0$, so incrementing l_k does not violate any constraint. This follows from Lemma 16 since $s_k > s_{k-1} \geq 0$ implies $s_k \geq 1$.

2. **Injectivity:** Different triples at weight w produce different targets at weight $w + 1$. This requires showing that the map $s \mapsto s'$ (where $s' = s + e_{k-1} - e_k$ in the cyclic notation) is injective on the set of slack vectors appearing at weight w. Since the slack determines the triple uniquely (given the weight), and the slack transformation is determined by the rule, it suffices to show that the induced map on slacks is injective. This amounts to verifying that no two distinct slack vectors map to the same successor slack under the cyclic descent rule.

The injectivity in point (2) appears to be the key obstacle. Computer verification confirms it for all tested values, but a uniform combinatorial argument remains to be found. The cyclic symmetry $\mathbb{Z}/3\mathbb{Z}$ of the slack simplex (rotating indices $1 \to 2 \to 3 \to 1$) plays a central role: the rule respects this symmetry in the sense that it "follows the cyclic descent", and the $3 \nmid d$ condition ensures the absence of a fixed point under the cyclic action.

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